

For star Conn.

$$I_L = I_p$$

$$V_L = \sqrt{3} V_p$$

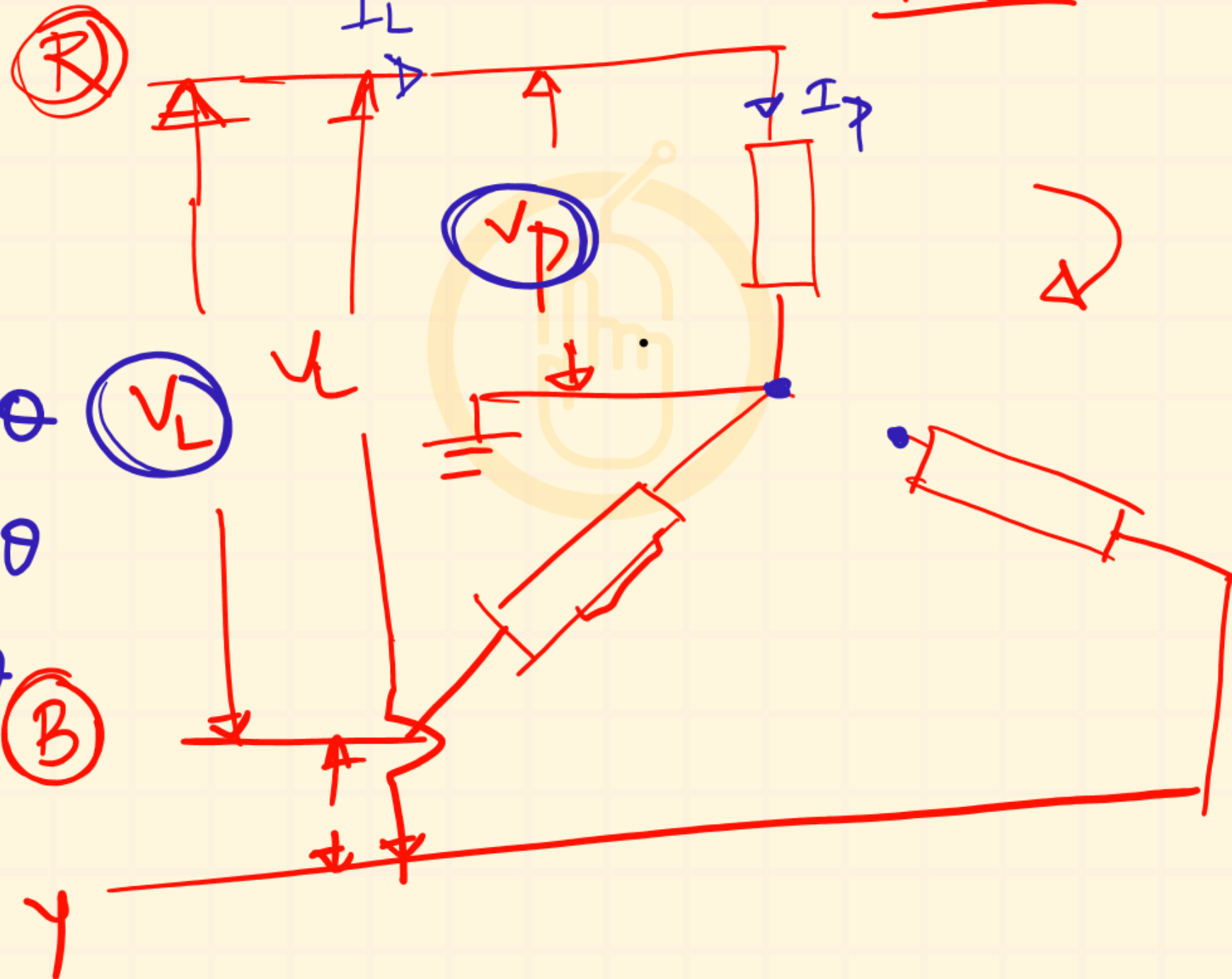
$$P = \sqrt{3} V_L I_L \cos \theta$$

$$Q = \sqrt{3} V_L I_L \sin \theta$$

$$\phi = 2 V_L I_L \cos \theta$$

poly-phase

star

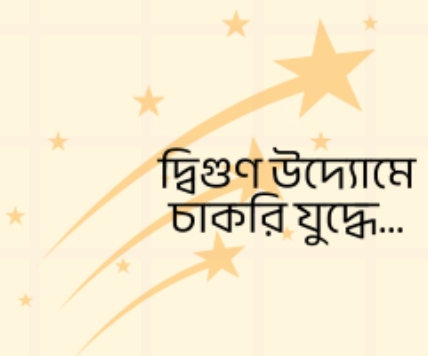


3-φ

MCO

R Y B R B Y

$\downarrow + (V_L)$



দ্বিগুণ উদ্যোগে
চাকরি যুদ্ধে...

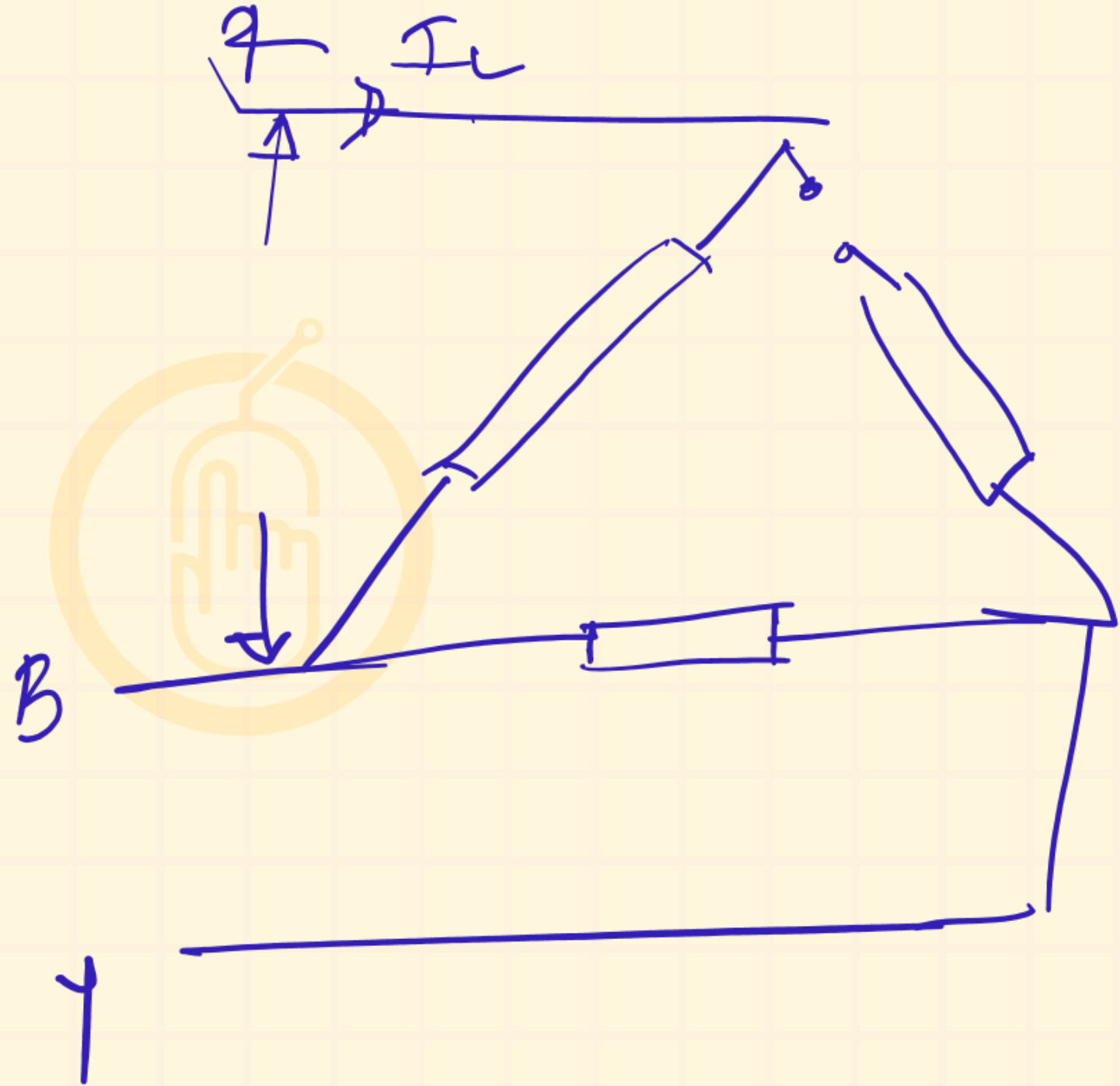
$$I_L = \sqrt{3} I_p$$

$$V_L = V_p$$

$$P = 3 V_p I_p \cos \theta$$

$$Q = 3 V_p I_p \sin \theta$$

$$P = 2 V_p I_p \cos \theta$$



* তার - ϕ Delta

Total power - ϕ কত বেশ

কমবে?

$\phi = \frac{P_T}{3}$

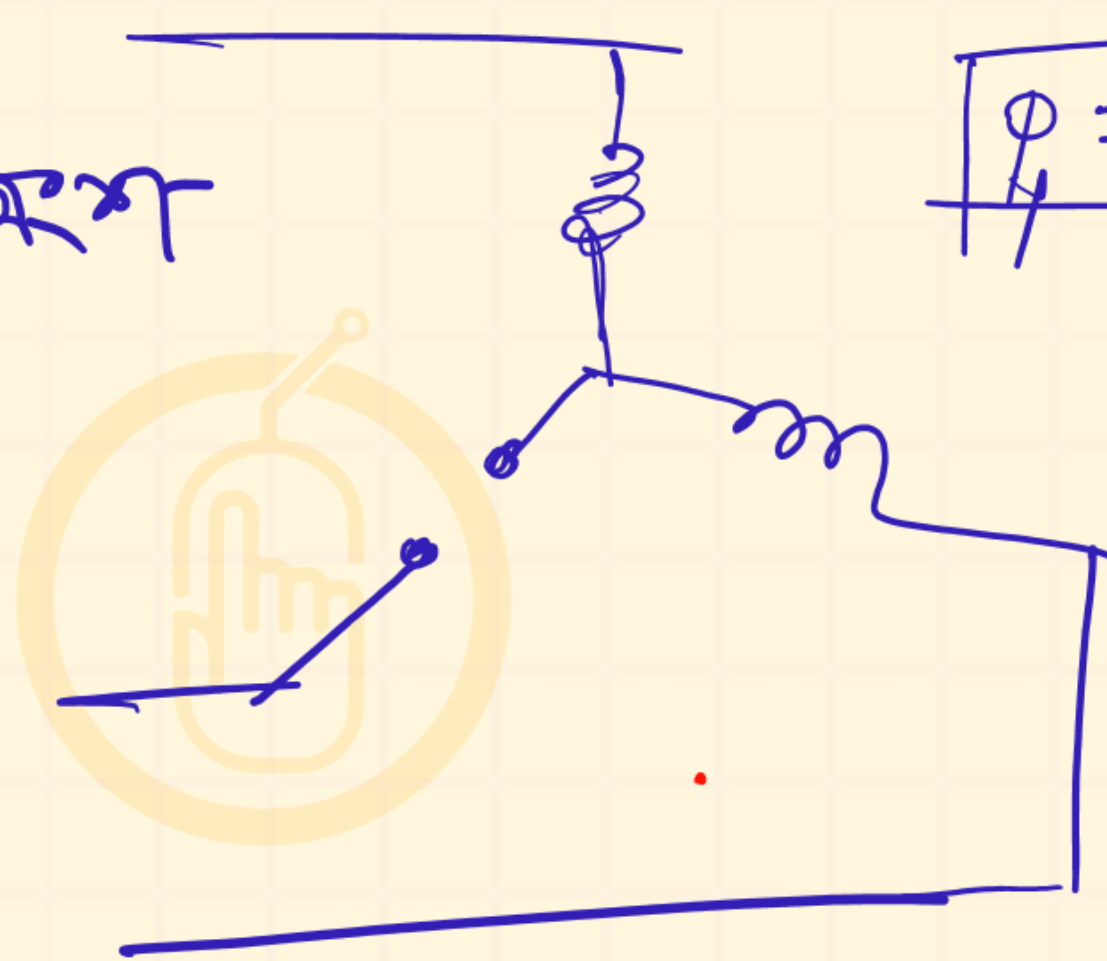
$$P_T = 3 I^2 R$$

$$I^2 R = \frac{P_T}{3}$$

$\phi = \frac{P_T}{3}$

$$P = 2 I^2 R$$

$$= \frac{2 P_T}{3}$$



$$\phi = 3 I^2 R$$

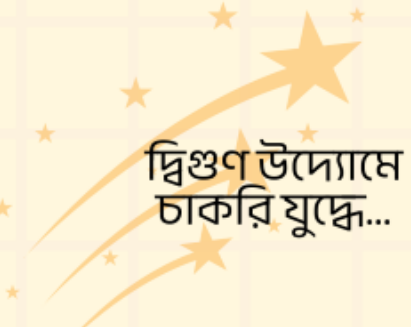
$$2 V_L I_L \cos \phi$$

$$2 I^2 R$$

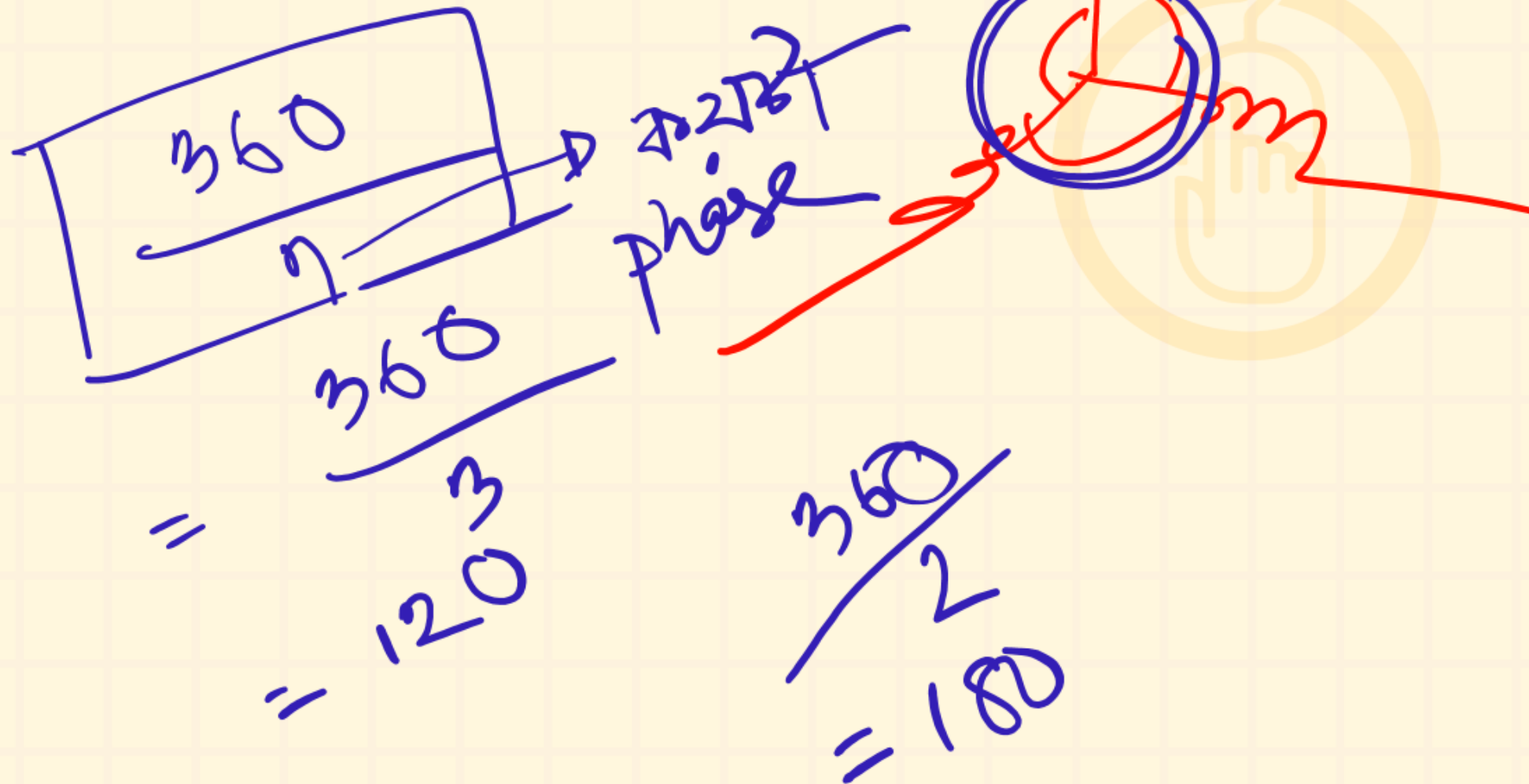
$$P_T = \frac{2 \phi}{3}$$

$$= \frac{3 P_T - 2 P_T}{3} = \frac{P_T}{3}$$

$$= \frac{1}{3} P_T$$



দ্বিগুণ উদ্যোগে
চাকরি যুদ্ধে...

 60°

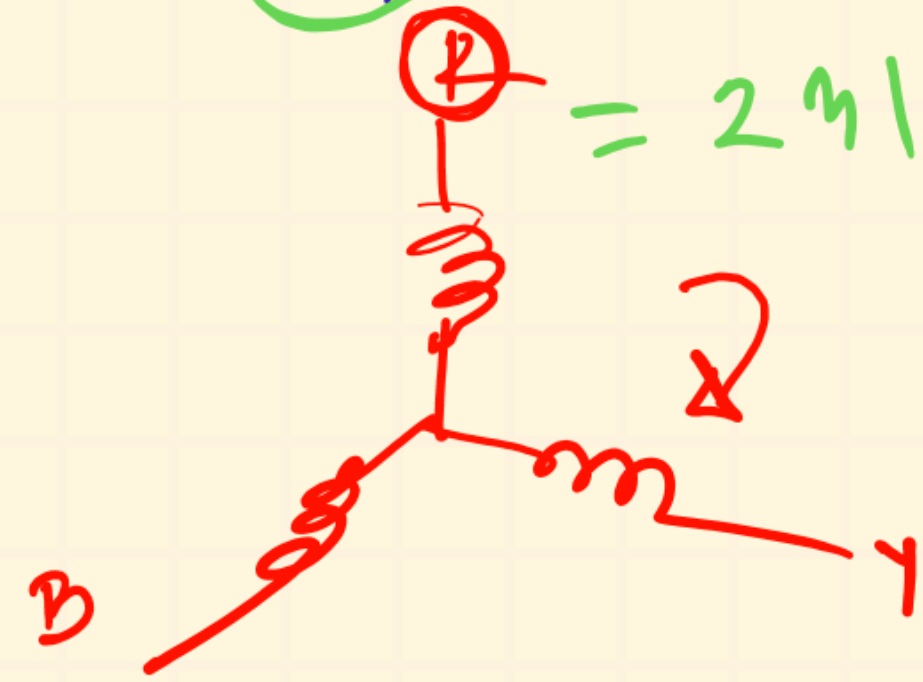
3-4

$I_R = 23 \angle 0$

$I_Y = 23 \angle -120^\circ$

$I_B = 23 \angle 120$

$I_{\phi} = 23 \angle -240$



+ (ve)
- (ve)

no cca cca cca

no c

cca

no c

R Y B

no c no c

no c
R Y B

no c
R Y B

C.W $\rightarrow$ - (ve)
A.C.W $\rightarrow$ + (ve)

27B
↓
YBR

27B 27B
↓
t(vr)

* Wattmeter

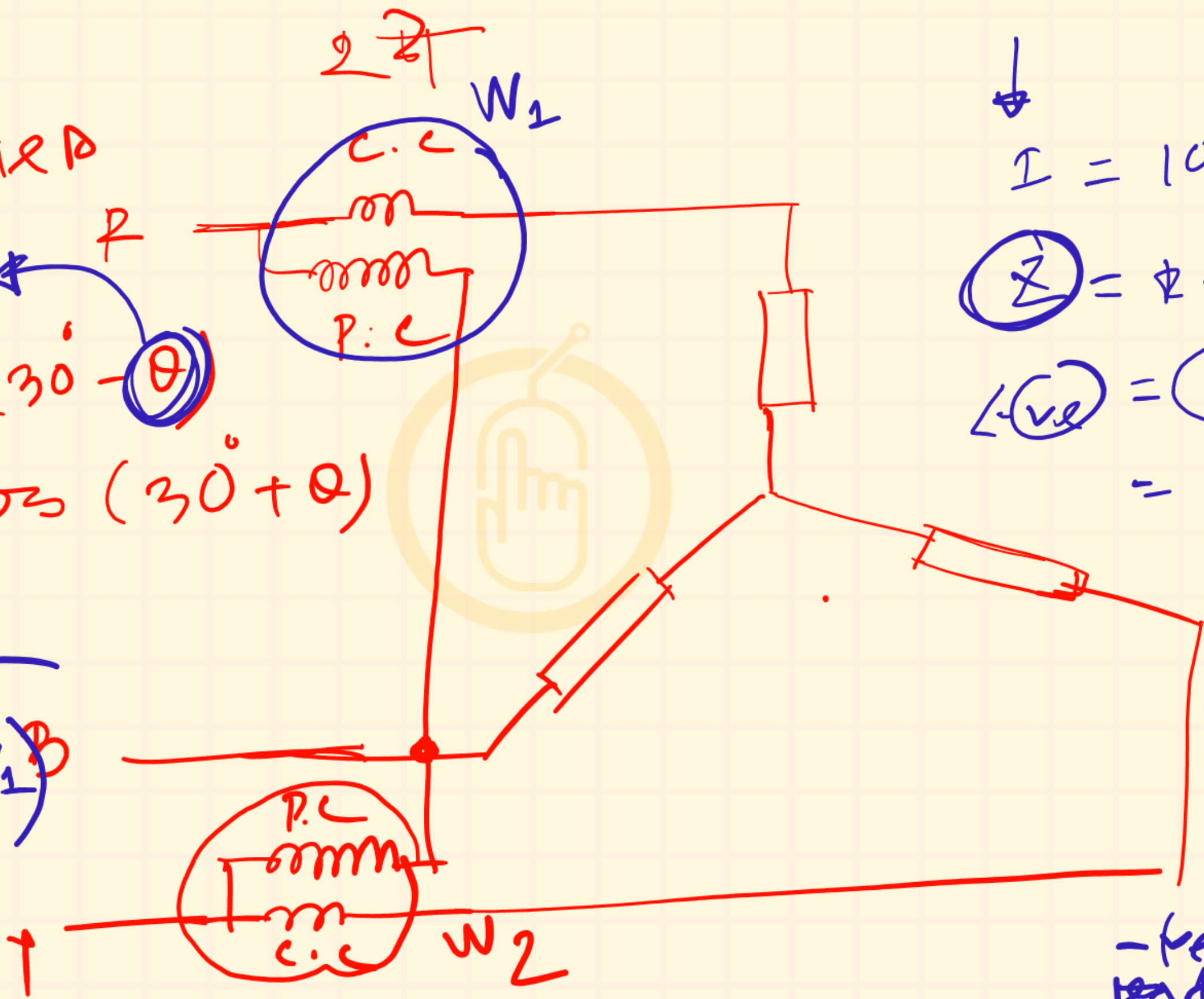
Z-PT

$$W_1 = \underline{V}_L \underline{I}_L \cos(\angle 0^\circ - \theta)$$

$$W_2 = \underline{V}_L \underline{I}_L \cos(\angle 0^\circ + \theta)$$

$$P_T = W_1 + W_2$$

$$Q_T = \sqrt{3}(W_2 - W_1)$$



$$I = 10 \angle 30^\circ$$

$$\underline{Z} = R + j(X_L - X_C)$$

$$\angle(V_L) = \angle(R - jX_C)$$

$$= R + jX_L$$

$\angle +ve$

Inductive
+ve
pf = 1
cap.

-ve
cap.

দ্বিগুণ উদ্যোগে
চাকরি যুদ্ধে...

$$\tan \theta = \frac{Q}{P} = \frac{\sqrt{3}(w_2 - w_1)}{w_2 + w_1}$$

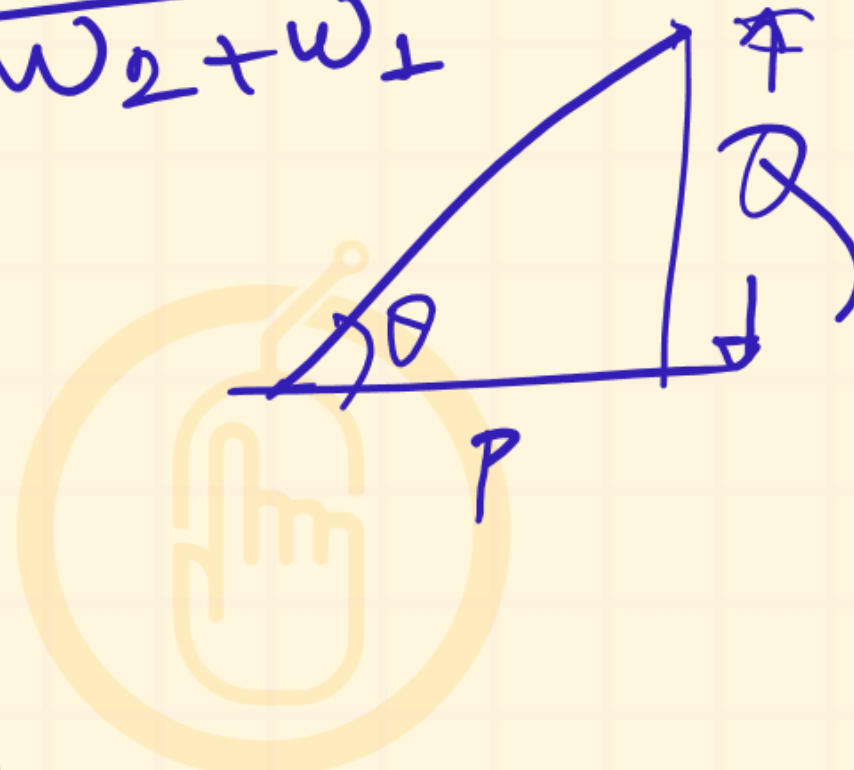
$w_2 > w_1$ (+ve) $\downarrow$ lag

$w_2 < w_1$ (-ve) $\downarrow$ lead

$$\theta = \tan^{-1} \left(\frac{Q_T}{P_T} \right)$$

$\downarrow$ = ()

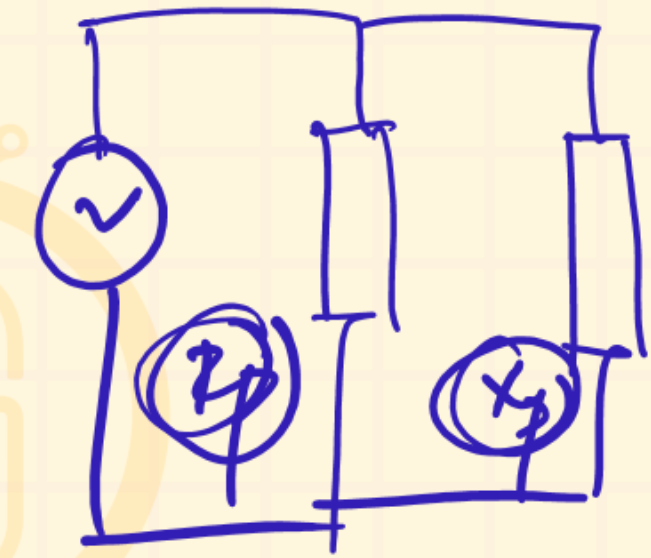
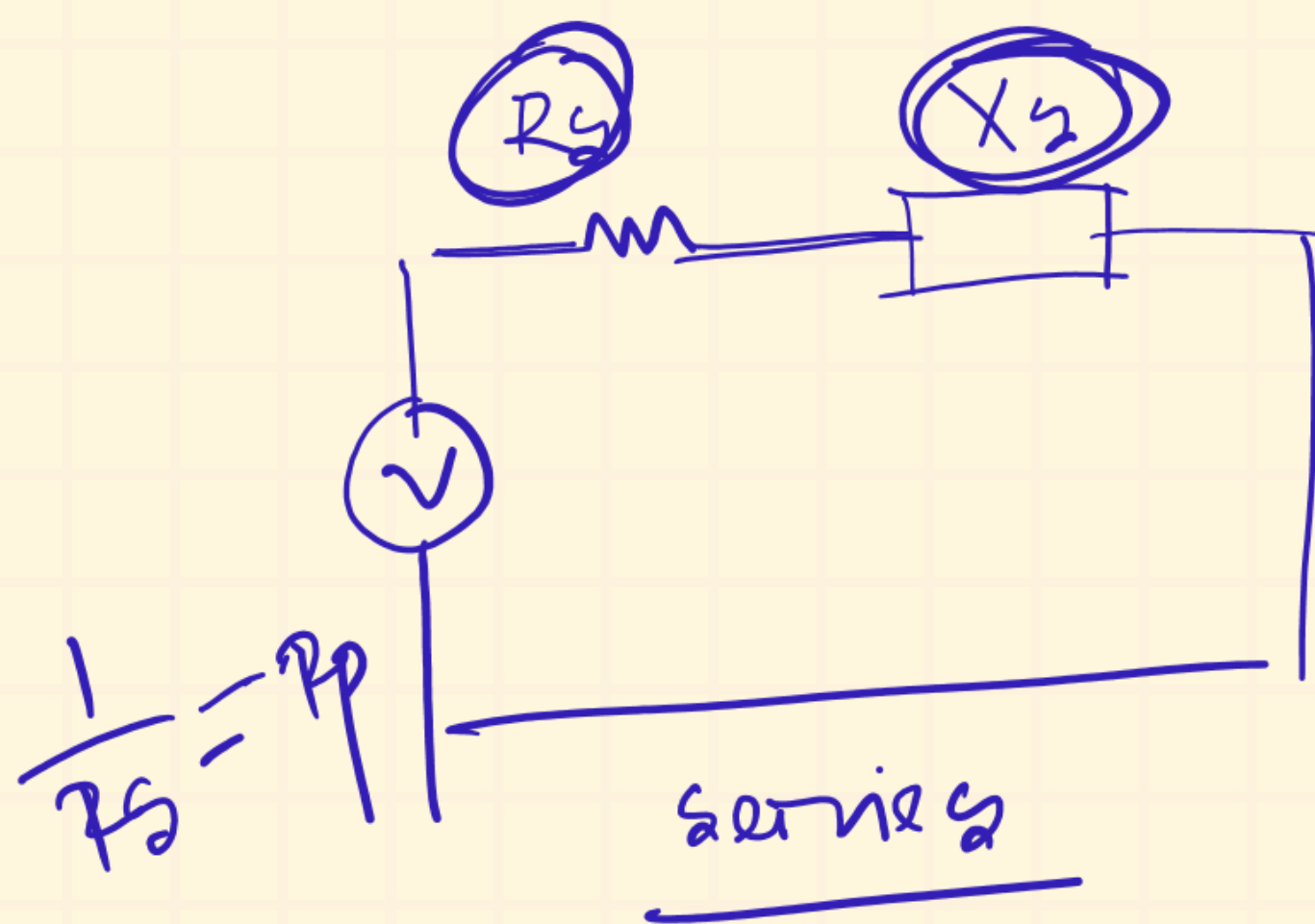
p. f = $\cos \theta$ $\downarrow$ lag



4
z

$$Z = \frac{1}{Y}$$

$$Y = R \pm jX$$



Wattmeter

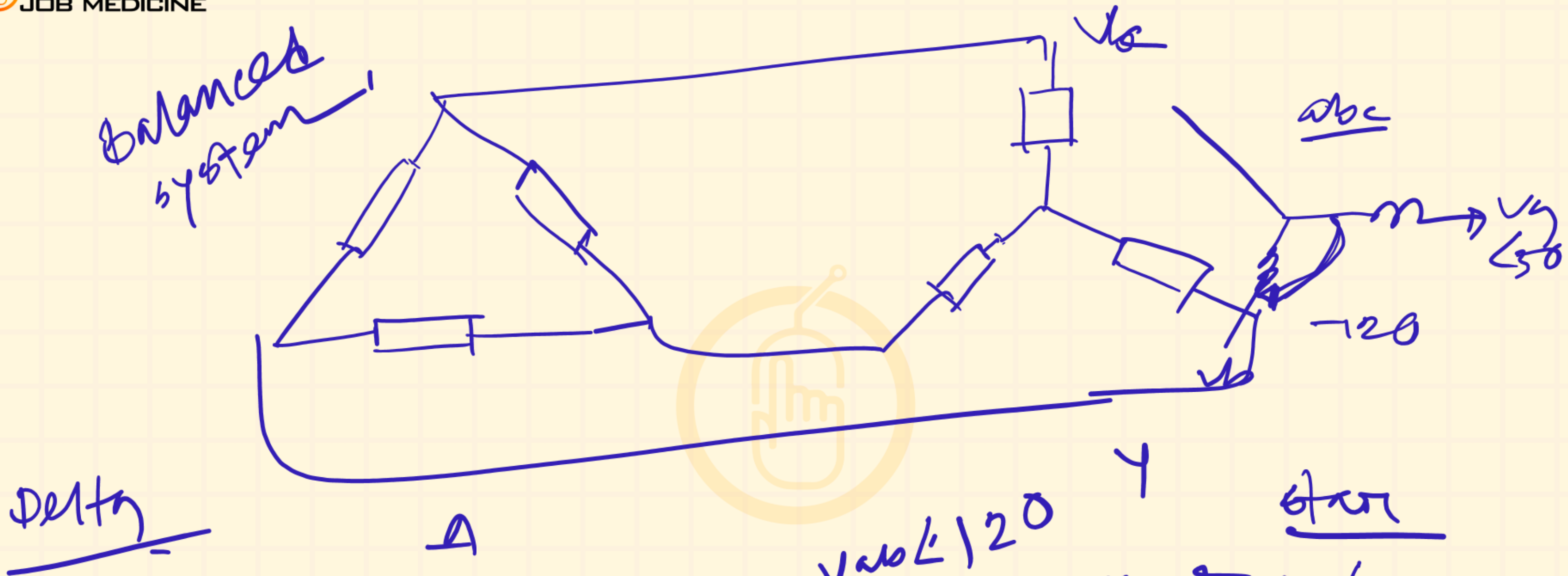
Teacher Book

$$Z_s = R_s + jX_s$$

$$Y = \frac{1}{Z_s} = G \pm jB$$

$$P = \frac{1}{S} = \frac{1}{P}$$

দ্বিগুণ উদ্যোগে
চাকরি যুক্ত...



$V_{an} = V_{ab} \angle 120^\circ$

$V_L = \sqrt{3} V_P \angle 30^\circ$

$I_L = \sqrt{3} I_P \angle 30^\circ$

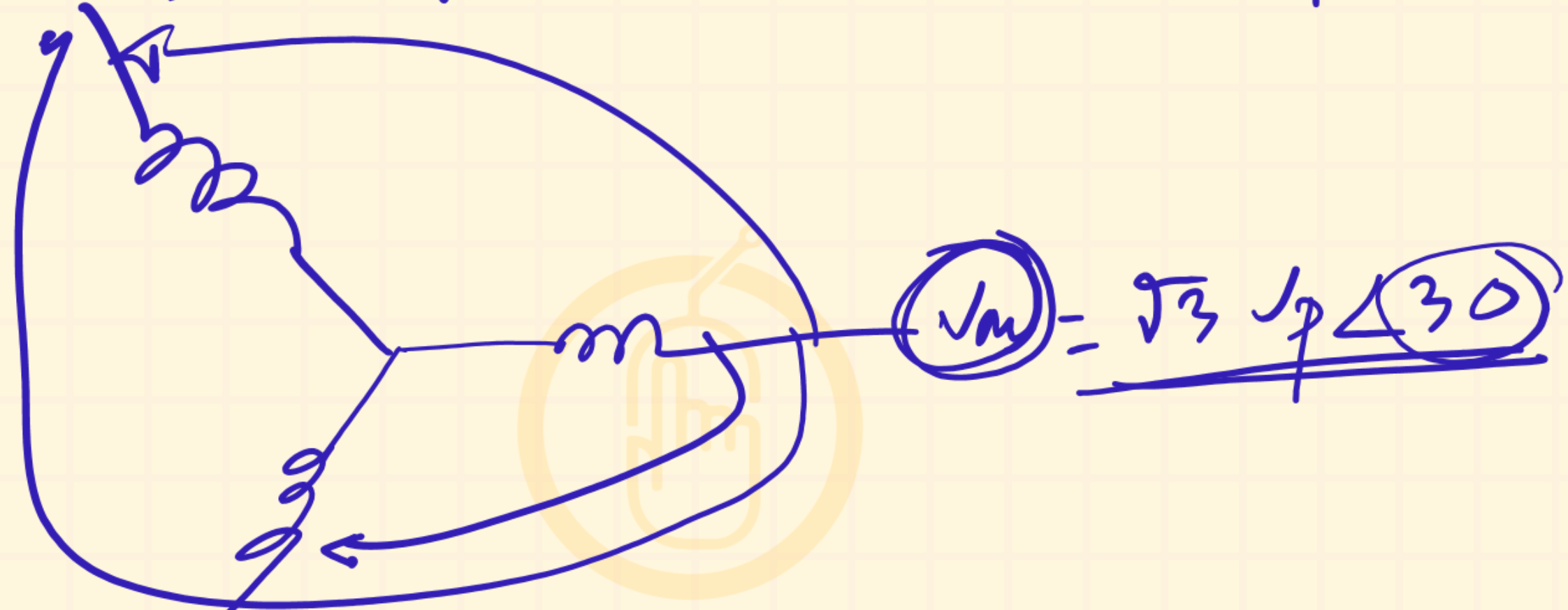
$V_{bc} = V_{ab} \angle -120^\circ$

$V_{ca} = V_{ab} \angle -240^\circ$

দ্বিগুণ উদ্যোগে
চাকরি যুদ্ধে...

$$V_a = V_m \angle -240^\circ = V_a \angle 120^\circ$$

$$V_{ab} = \sqrt{2} V_p \angle 30^\circ$$



$$\underline{V_m = \sqrt{3} V_p \angle 30^\circ}$$

$$V_a = V_{ab} \angle -120^\circ$$

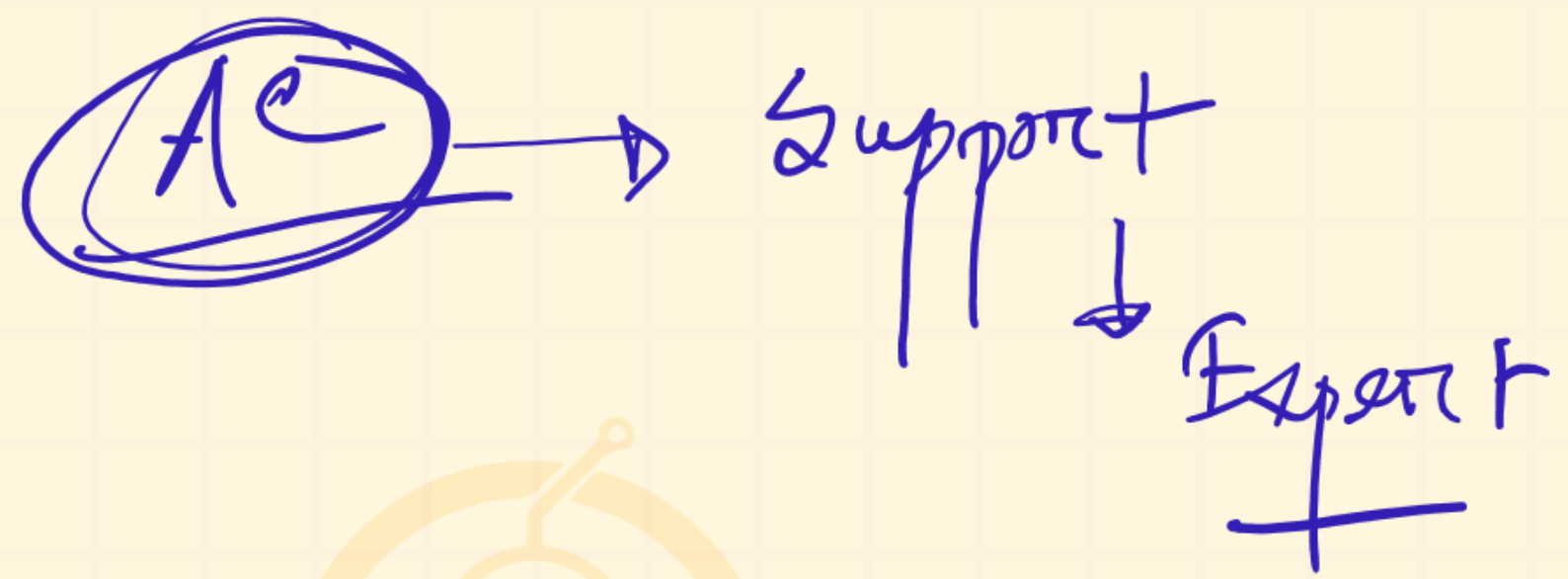
$\textcircled{*} I_L = \sqrt{3} I_p \angle 30^\circ$

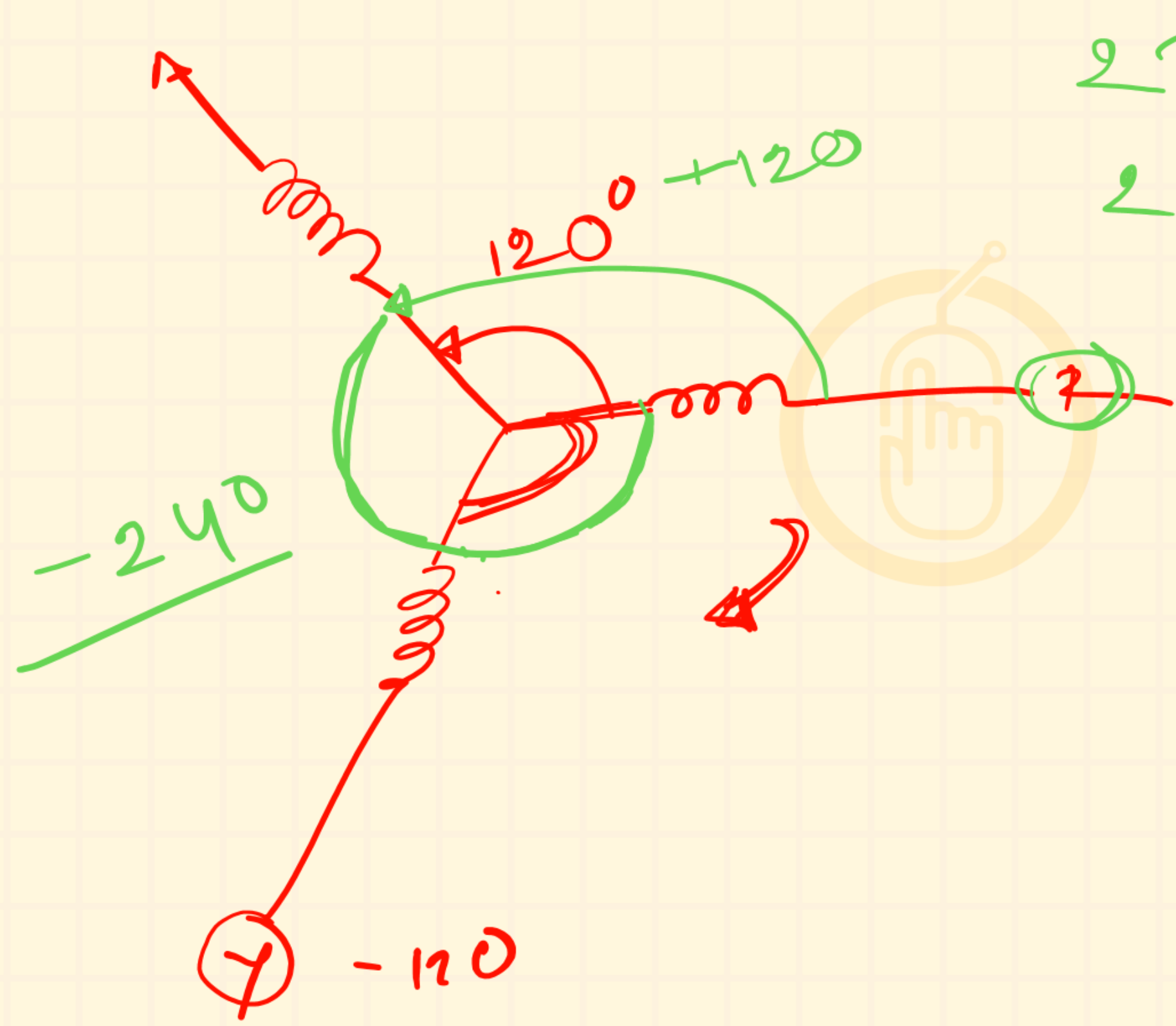


$\therefore \boxed{I_a = \sqrt{3} I_p \angle -30^\circ}$

$I_b = I_a \angle -120^\circ$

$I_c = I_a \angle 240^\circ$





$231 \angle -240$
 $231 \angle 120$

$E_R = 231 \angle 0$
 $E_R = 231 \angle -120^\circ$

120°

-240